Astronomical Bycatch: progress report

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1 Summary

The aim is to get light curves for the stars in archival images using the new extraction API in our private EXOTIC fork (`exotic.extract`). This report covers the first day of that work:

- Code to search and download raw data from the ESO Science Archive, and to find archive sequences that cover a known transit.
- A pipeline that plate-solves frames locally against Gaia DR3, runs `extract_sequence` on every usable star in the field, fits the target's transit and writes all light curves to JSON.
- **Validation:** on the HAT-P-32 b MicroObservatory night, the fitted mid-transit time agrees with EXOTIC's own validation result to within $\sim 1.5\sigma$.
- A wide-field ESO test night (POTS-1 b, ESO/MPG 2.2 m WFI) is downloaded, calibrated and plate-solved. Extraction is running.

2 Data sources

ESO Science Archive (raw frames). Metadata was queried through ESO's catalogue query service, and transiting planets through the NASA Exoplanet Archive. The two optical wide-field imagers, WFI and OmegaCAM, have 182 long single-pointing sequences between them (at least 40 frames over at least 2 hours). Only one catches a known transit: **POTS-1 b, WFI, night of 2007-05-10**. It has 310 Rc-band frames of 26 s over 8.2 h, covering the full transit with $\sim$2.5 h of data before and 2.8 h after. The night has 24 biases, 4 twilight flats and 10 screen flats; the night is 46 GB decompressed. POTS-1 is faint ($G = 17.3$) but its transit is deep, and the field is crowded with background stars.

MicroObservatory frames. A collaborator's repository of raw MicroObservatory frames covering seven hot-Jupiter targets. The HAT-P-32 night is byte-identical to EXOTIC's own sample data, which makes it a direct check against EXOTIC.

3 Code and storage

`access/` Searches the ESO archive, matches sequences to known transits, and downloads a target's frames and calibrations (`python -m access.fetch targets/<t>.json`). Downloads can be restarted. Each target is a small JSON file in `targets/`.

`pipeline/astrometry.py` Local plate solving against Gaia DR3, so no image is uploaded to an external service. Raw WFI and MicroObservatory frames carry no sky coordinates, and the MicroObservatory header pointing can be $\sim 0.3^\circ$ off.

`pipeline/microobs.py`, `pipeline/wfi.py` Pipelines for each instrument. WFI handles 8 detectors per frame: overscan trimming, master bias and flat, and a separate plate solve per detector.

`pipeline/store.py` JSON results, one file per sequence (and per detector). Each holds the observation details, per-frame times and airmass, and a light curve for every star keyed by Gaia DR3 ID. The layout maps onto the existing kdb+ schema (`sources`, `obs_sessions`, `observations`).

Data and results stay on the department compute server; nothing large is kept on laptops. Code is on the `huw-leggate/claude-md` branch of `creanero/bycatch`.

4 Results

4.1 HAT-P-32 b: validation

142 frames of 60 s. The plate solve matched 184 Gaia stars to 1.5 arcsec, and 83 stars were photometered. The target's light curve is its flux divided by the summed flux of the comparison stars. It was fitted with EXOTIC's least-squares `fit_transit`, with a linear airmass trend removed. Three parameters are free: the mid-transit time T_{mid}, the planet-to-star radius ratio R_p/R_s, and a flux scale. R_p/R_s is the planet's radius divided by its host star's radius, which sets the transit depth (depth $\approx (R_p/R_s)^2$). The orbit (period, a/R_s, inclination) and the limb darkening are held fixed. Table 1 gives the results for three aperture sizes, and Figure 1 shows the light curve.

Table 1: HAT-P-32 b transit fits for three photometric aperture radii, given as multiples of the measured stellar FWHM. *Scatter* is the point-to-point scatter of the relative light curve. T_{mid} is given relative to EXOTIC's own result for this night (2458107.71380 $\mathrm{BJD_{TDB}}$). R_p/R_s is the fitted planet-to-star radius ratio; the published value is 0.149.

Aperture	Radius (px)	Scatter	T_{mid} offset (min)	Fitted R_p/R_s
1.0×FWHM	2.5	0.53%	-0.6 ± 2.6	0.151 ± 0.003
1.4×FWHM	3.5	0.48%	$+2.1 \pm 2.4$	0.157 ± 0.003
1.9×FWHM	4.7	0.57%	$+4.0 \pm 2.8$	0.156 ± 0.004

All three mid-transit times are within $\sim 1.5\sigma$ of EXOTIC's result. The fitted R_p/R_s runs 1–5% above the published 0.149, probably because the limb darkening is fixed at approximate values; a depth error doesn't move the timing.

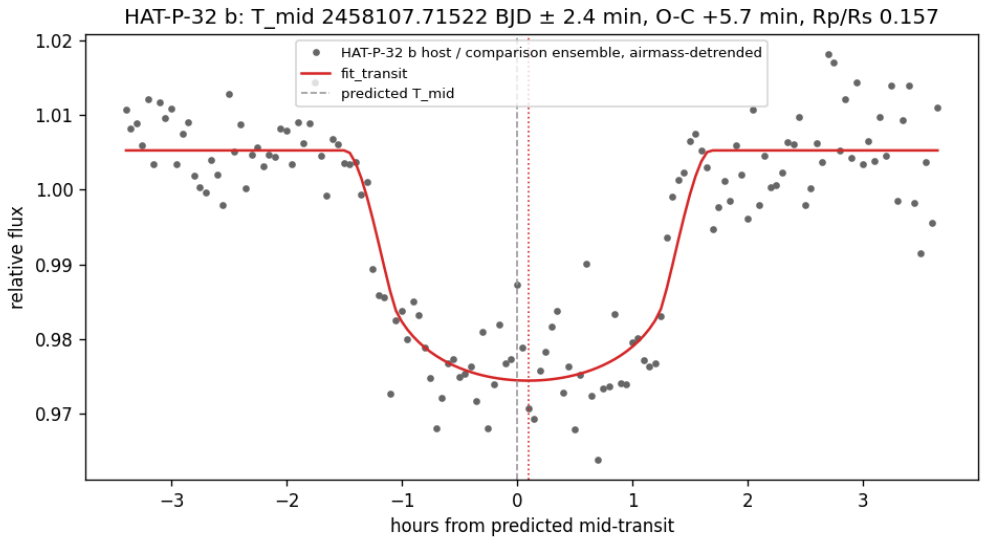


Figure 1: HAT-P-32 b relative light curve (grey points) with the fitted transit model (red), using the 1.4×FWHM aperture. The x-axis is hours from the predicted mid-transit time. The fitted T_{mid} is 5.7 min after the ephemeris prediction and 2.1 min after EXOTIC's result (Table 1).

4.2 WASP-36 b

55 frames of 60 s. Plate solving and frame alignment worked, but the target ($G = 12.6$) is limited by sky noise: the best scatter is 2.3% per point, with a 1.0×FWHM aperture. The transit is ∼1.8% deep, and a 66-minute gap leaves only 3 frames before ingress. The fit (Figure 2) puts T_{mid} $+26 \pm 43$ min from the prediction and $R_p/R_s = 0.09 \pm 0.06$ (published 0.134). Both are consistent with expectations, but too uncertain to be useful.

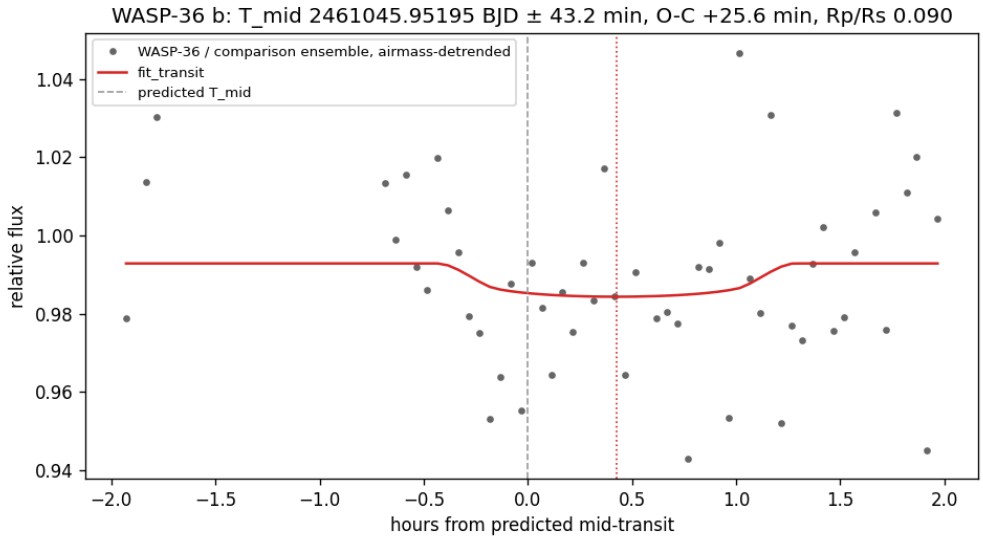


Figure 2: WASP-36 b relative light curve (grey points) with the fitted transit model (red), using the 1.0×FWHM aperture. The dashed grey line is the predicted mid-transit time and the dotted red line the fitted one. The scatter is comparable to the transit depth.

4.3 POTS-1 b, WFI (in progress)

- **Calibration:** gain and read noise were measured for each detector from bias and flat pairs, because the headers record gain for only two detectors and read noise for none. Gain came out at 1.94–2.28 e^-/ADU and read noise at 5.6–6.2 e^-, consistent with WFI's documented values.
- **Astrometry:** all 8 detectors plate-solved, with 36–163 Gaia stars each and residuals of 0.06–0.11 arcsec. POTS-1 is on chip 3.
- **Extraction:** 801 stars down to $G = 18.5$ across all 310 frames, split over 16 processes. Results to follow.

5 Issues and next steps

- **Speed:** centroid fitting in `extract_frame` takes ∼70 ms per star per frame, which dominates run time (about 55 s per WFI frame for 800 stars). Running in parallel works around it for now. This is a useful benchmark for the GPU work in EXOTIC.
- **Aperture size:** the default FWHM-scaled aperture is too large for faint, sky-limited stars. It should be tuned per star, or chosen by sweeping, for bycatch at scale.
- **Limb darkening:** fixed approximate coefficients bias the fitted R_p/R_s. Timing is unaffected.
- **Next:** finish POTS-1; run the remaining MicroObservatory targets; look for variability among the field stars in the JSON results; then scale the search to OmegaCAM and the other WFI sequences.